



Project and Professionalism

6CS007

Dart Report

Medical Lab Management System

Clinical

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1. Introduction

Diagnostic labs are essential to Nepal's healthcare system since they offer vital tests for illness identification, treatment choices, and patient tracking. Many labs in Nepal, nevertheless, continue to use manual procedures such paper-based registration, handwritten reports, and in-person doctor consultations. These antiquated procedures frequently result in delays, mistakes in the processing of data, and problems for patients who might have to travel great distances or wait for hours to obtain their medical advice and reports.

By creating a web-based platform that digitizes and unifies the entire laboratory process, the proposed project, "Clinical – Medical Lab Management System," seeks to address these problems. With their Citizenship ID, patients in Nepal will be able to register, schedule tests online, pay at the lab, view their reports digitally, and consult doctors without needless effort. While administrators can keep an eye on things using dashboards and analytics, lab workers can effectively handle samples and produce reliable reports. Nepali healthcare may become more patient-friendly, transparent, and efficient by putting such a system into place. It will also improve accessibility, especially in distant places, and reduce errors.

1.1. Aims

The main aim of this project is to build a simple and reliable web-based system that helps medical labs in Nepal manage patients, tests, reports, and doctor consultations in one place. It seeks to make the whole process faster, more accurate, and easier for both patients and healthcare providers.

1.2. Objectives

- To build an easy-to-use web system for patients to register, book tests, and track their reports in Nepal.
- To allow patients to make payments and manage sample collection smoothly online.
- To help lab technicians record, verify, and publish accurate test results efficiently.
- To enable doctors to access patient reports and share consultation notes digitally.
- To provide an admin dashboard for monitoring lab activities, patient flow, and feedback effectively.

1.3. Artefacts

1.3.1. Functional Decomposition Diagram

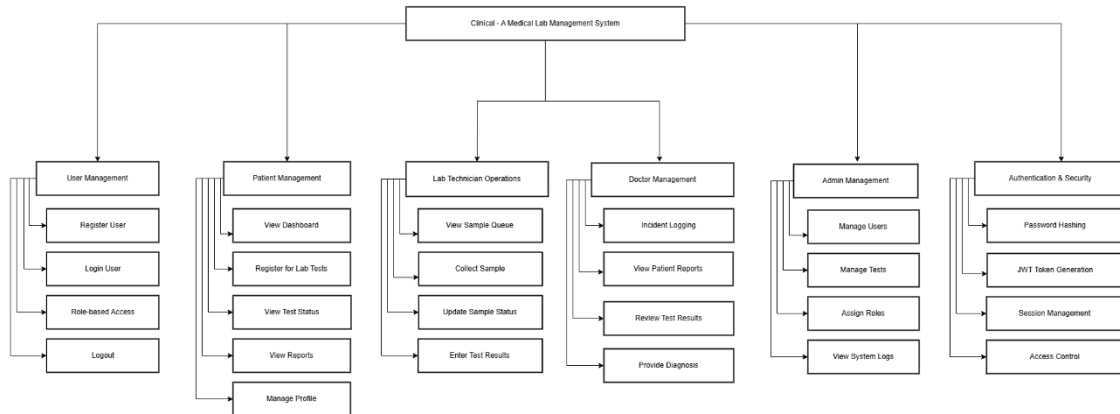


Figure 1 Functional Decomposition Diagram

1.3.2. Work Breakdown Structure

User Management

Handles user-related operations:

- Register User: Allows new users to sign up.
- Login User: Enables users to log into the system.
- Role-based Access: Grants permissions based on user roles (e.g., patient, doctor, admin).
- Logout: Ends the user session.

Patient Management

Manage patient activities:

- View Dashboard: Displays patient-specific information.
- Register for Lab Tests: Allows patients to book tests.
- View Test Status: Shows progress of requested tests.
- View Reports: Provides access to test results.
- Manage Profile: Lets patients update personal details.

Lab Technician Operations

Handles lab-related tasks:

- View Sample Queue: Shows pending samples to process.
- Collect Sample: Records sample collection details.
- Update Sample Status: Tracks sample progress.
- Enter Test Results: Inputs test outcomes into the system.

Doctor Management

Supports doctors in managing patient care:

- Incident Logging: Records of medical incidents or observations.
- View Patient Reports: Accesses patient test reports.
- Review Test Results: Evaluates lab results.
- Provide Diagnosis: Adds diagnosis based on test results.

Admin Management

Controls administrative functions:

- Manage Users: Adds or removes users.
- Manage Tests: Updates test types and details.
- Assign Roles: Sets user roles (doctor, technician, patient).
- View System Logs: Monitors system activities for security and auditing.

Authentication & Security

Ensure secure access and data protection:

- Password Hashing: Secures user passwords.
- JWT Token Generation: Provides secure authentication tokens.
- Session Management: Handles user sessions.
- Access Control: Restricts access based on roles and permissions.

1.4. Scope and Limitation of the Project

1.4.1 Scope

The scope of this project is to develop a web-based Laboratory Management System that helps streamline the process of booking and managing laboratory tests. The system allows different types of users such as patients, laboratory technicians, and administrators to interact with the platform according to their roles. Patients can view available medical tests, book tests online, and track the status of their reports. Laboratory technicians can manage the test workflow by updating stages such as sample collection, processing, and publishing results. Administrators can manage users, maintain the test catalog, and monitor overall system activities.

The system focuses on improving efficiency in laboratory operations by reducing manual paperwork and providing a centralized platform for managing patient test records. It also enables administrators to monitor patient bookings and technician activities in real time. Overall, the project aims to create a simple, organized, and user-friendly system that supports basic clinical laboratory operations digitally.

1.4.2 Limitations

- The system will focus only on medical lab management and will not cover hospital-wide management like pharmacy, inpatient care, or billing beyond lab tests.
- Online payment integration may be limited to basic gateways initially, without advanced features like international payments.
- Some features like automated test imports from lab instruments or multi-language support may be optional and not fully implemented in the first version.
- As a solo developer, complex AI-based analysis or advanced analytics will not be included.

2. Literature Review

2.1. Introduction

Medical laboratories sit at the core of modern healthcare: they confirm diagnoses, guide therapy, and track outcomes. Yet in many contexts, especially resource-constrained settings labs still depend on paper forms, ad-hoc spreadsheets, and fragmented software, a combination that slows turnaround, invites transcription error, and complicates quality assurance. Contemporary Lab Information/Lab Management Systems (LIS/LIMS) attempt to solve this by digitizing sample life cycles end-to-end (registration, accessioning, testing, validation, reporting), coupling these workflows to inventory, quality control, and analytics, and increasingly to telemedicine/consultation and secure patient reporting. Recent literature spans (i) conceptual and managerial guidance for running safe, compliant laboratories; (ii) engineering proposals for end to-end LIS/LIMS with an emphasis on architecture, security, and scalability; and (iii) market/platform analyses describing what real systems provide. Together they suggest a convergence on role-aware, workflow-centric, analytics-enabled systems that reduce error, accelerate turnaround time (TAT), and strengthen auditability across pre-analytic, analytic, and post-analytic phases (Le, N.T., Thwe Chit, M.M., Truong, T.L., Siritantikorn, A., Kongruttanachok, N., Asdornwised, W., Chaitusaney, S. & Benjapolakul, W). Evidence from health-IT security research adds another pressure: health data breaches remain frequent and costly, making privacy by design and tamper-resistant audit trails critical features for any lab platform.

2.2. Research Paper

2.2.1. Smart specimen transport and RFID/NB-IoT for Sample Tracking

Le et al. (2023) present a practical prototype that integrates passive RFID tagging of specimen tubes with an NB-IoT enabled smart transport box to monitor environmental conditions (temperature, humidity) and GPS location during specimen transfer. Their pilot at a large Thai hospital shows RFID's advantages over barcodes for non-line-of-sight, multi-read operations and demonstrates how continuous environmental telemetry can reduce pre-analytical errors and lost samples two of the biggest sources of lab error (Le et al., 2023). The study also argues that reusable waterproof RFID tags reduce running costs and that cloud logging of transport telemetry enables audit trails and earlier detection of compromised specimens. For laboratories in low resource settings, the paper suggests that RFID+IoT solutions can meaningfully reduce specimen misplacement, speed accessioning, and provide evidence to triage samples that experienced unacceptable temperature excursions. Limitations noted include costs of infrastructure, the need for integration with existing LIMS, and pilot-scale testing rather than long-term deployment. Overall, the work demonstrates a technically feasible path to improve sample chain-of-custody and pre-analytical quality in hospital labs.

2.2.2. RFID adoption and real-world healthcare lessons

Systematic and review studies on RFID adoption in healthcare (Abugabah et al., 2023) synthesize field experiences and highlight recurring facilitators and barriers. RFID improves sample and asset tracking, reduces manual scanning time, and supports automation of workflows; however, real world adoption is often slowed by integration challenges with laboratory information systems, concerns about data privacy, tag/read reliability in crowded radio environments, and upfront costs (Abugabah, 2023). Reviews emphasize that technology alone is insufficient successful deployments require process redesign, staff training, and clear ROI metrics (e.g., reductions in sample losses, turnaround time, and repeat tests). For lab managers and project leaders, the lesson is to adopt a phased approach: begin with high-risk flows (e.g., inter-facility transport), validate read rates, then extend RFID coverage while ensuring LIMS and dashboards reflect new data streams. This layered approach aligns well with project's objectives (sample tracking, audit logs, admin analytics) and suggests a roadmap for pilot → integration → scale.

2.2.3. Patient portals and direct access to lab results, benefits and risks

Digital patient portals that provide direct access to laboratory results have been widely studied since 2021. Systematic reviews (Carini et al., 2021; Steitz et al., 2025) show that patient portals improve engagement, support patient understanding of health status, and can speed follow-up actions; features most consistently valued are direct lab result access and secure messaging with clinicians (Carini, 2021). However, studies also report potential harm: patients accessing sensitive or abnormal results without clinician commentary may experience anxiety or contact clinicians urgently, increasing messaging loads. To balance benefits and harms, the literature recommends: (1) contextualized result displays (reference ranges, plain-language explanations), (2) delayed release windows for highly sensitive tests where appropriate, (3) in-portal clinician comments or interpretation tools, and (4) analytics on portal usage to allocate clinician resources. For Clinical system, integrating lab PDF reports with an explanatory layer and a secure messaging or triage workflow will maximize the portal's positive impact while guarding against increased clinician burden.

2.2.4. Integration technologies, why LIMS/LIS interoperability matters

Recent work on software-to-software (S2S) integration in laboratory contexts (Edayan, 2024) stresses that the clinical value of any digital lab system depends on robust interoperability: LIMS/LIS must exchange orders, results, and specimen metadata, and audit logs with EHRs, billing, and analytics platforms. The literature highlights standards-based approaches (HL7 FHIR, HL7 v2 for legacy labs, LOINC for test codes) and middleware patterns (message brokers, API gateways) that simplify connecting heterogeneous systems. Integration improves patient safety (reduces transcription errors), speeds reporting, and enables aggregated dashboards for administrators to achieve all objectives in proposals. Major obstacles documented include mapping local test codes to standards, handling asynchronous result flows (partial results; amended reports), and ensuring secure, auditable exchanges that meet legal/privacy requirements. The implication for the Clinical MERN architecture is clear: design

backend APIs compatibility in mind, include robust logging for audit and error reconciliation, and adopt standard code sets early to avoid costly reworking later.

2.2.5. LIMS / LIS impacts on efficiency, quality and low-resource settings

More focused reviews and case studies from 2022–2024 (Cereus, 2024) show that LIMS deployments consistently improve laboratory turnaround times, reduce manual transcription errors, and facilitate quality assurance workflows when implemented with appropriate training and governance (Cureus, 2024). In low-resource settings, papers stress configurable, lightweight LIMS deployments that prioritize critical flows (sample accessioning, test result entry/verification, PDF report generation, and minimal offline capabilities). Implementation case studies note that even modest automation digitizing registration, using barcode/RFID labels for samples, and building a simple patient portal yields measurable improvements in report delivery and patient satisfaction. The reviews also warn against overly ambitious scope for first releases; staged rollouts with clear UAT and staff champions produce better adaptation. These lessons dovetail with my proposal’s stated scope (lab-centric system, phased features, admin dashboards) and support choosing a modular MERN architecture that can integrate with external services (payments, messaging) as later phases.

2.3. Systematic Review: Similar Applications and Ecosystems

2.3.1. Mobile clinical-guideline apps in Nepal: the e-mhGAP case

In Nepal, mobile applications have been adopted to support primary health care workers in applying clinical guidelines. The study by e-mhGAP (version of the WHO mhGAP Intervention Guide in app form) in rural districts (Chitwan and Ramechhap) found high levels of interest and positive attitudes from medical officers and prescribing health workers toward using the app-based clinical guideline. The app facilitated more uniform diagnosis and management of mental disorders across facilities, helped verify assessment flows, and reinforced learning of the guideline (Pokhrel et al., 2021). However, major barriers emerged: unreliable internet connectivity in remote health facilities, older health workers’ lower comfort with smartphones, potential patient perception that using an app during consultation implies inexperience, and duplication of workload when paper and mobile records both had to be maintained. A follow-up study describing the development of the e-mhGAP app prototype noted technical issues (slow navigation, UI/UX difficulties) in addition to infrastructure constraints (Luitel et al., 2023). For my project (lab-management system), the lessons are relevant: any mobile-app component (for patients, lab staff, doctors) will need offline or low-connectivity capability, intuitive UI (especially for less tech-savvy users), integration to minimize duplicate data entry, and attention to user perception (so using the app does not undermine trust).

2.3.2. Mobile apps for cancer-pain management and wider health service access in Nepal

Another domain of mobile apps in Nepal is for more specialized care. For example, the NAPCare PMG App (cancer pain management) was pilot-tested across four cancer-care centers in Nepal in 2022. Nurses and physicians used the app; about 48% reported being very familiar, but 21% 'very often' used apps for clinical care (Frontiers, 2022). This study highlights that even among specialized care settings, mobile apps face adoption hurdles e.g., training needs, workflow integration, and consistent usage. Although the app was viewed as useful, full integration into routine clinical workflows remains incremental. In parallel, the health-app ecosystem in Nepal more broadly (including telemedicine, pharmacy-delivery apps, health-information apps) shows rapid growth. For instance, the NepMeds App launched for pharmacy, health library and soon-to-be doctor consultations. However, a recent rapid review of digital health interventions in Nepal (Parajuli et al., 2022) found that many apps and digital solutions are constrained by infrastructural issues (electricity/internet), human-resource and policy gaps. For lab-management system, integrating a mobile-app component (patient access, sample-tracking notifications, doctor access) could leverage these examples but you should prioritize offline resilience, minimal data entry burden, and phased integration into existing workflows.

2.4. Conclusion

The development of the Clinical – Medical Lab Management System represents a significant step toward modernizing healthcare operations in Nepal. As the healthcare sector increasingly embraces digital transformation, this project directly addresses long-standing inefficiencies caused by manual and paper-based processes in diagnostic laboratories. The system aims to provide a centralized digital platform that integrates patient registration, test booking, sample tracking, report generation, and doctor consultations, ensuring greater transparency, accuracy, and efficiency throughout the diagnostic workflow. By leveraging the MERN (MongoDB, Express.js, React.js, Node.js) technology stack, the system ensures flexibility, scalability, and a responsive user experience. The platform also supports multiple user roles including patients, lab technicians, doctors, receptionists, and administrators to ensure smooth coordination between all stakeholders involved in the laboratory process. Furthermore, the inclusion of an admin dashboard and analytics provides decision-makers with real-time insights into patient flow, revenue generation, and service quality, contributing to better resource management and accountability. In the broader context, this project aligns with Nepal's growing emphasis on digital health solutions as outlined in the national e-Health Roadmap. It not only enhances service delivery but also lays the groundwork for integration with national systems such as DHIS2, promoting data driven decision-making and health system interoperability. The system's modular design also allows future integration of features like online payments, multilingual interfaces, and AI-based analytics to further strengthen healthcare service quality and accessibility. Although challenges such as internet reliability, user training, and data privacy must be addressed, these barriers present opportunities for iterative improvement and sustainable scaling. The project's adoption of agile methodology ensures adaptability, allowing continuous testing and refinement based on user feedback. Ultimately, the Clinical – Medical Lab Management System contributes to the vision of an inclusive, efficient, and technology-

driven healthcare ecosystem in Nepal. It offers a foundation upon which digital healthcare infrastructure can evolve improving diagnostic accuracy, reducing patient waiting times, minimizing human error, and making healthcare services more accessible to people across both urban and rural regions of the country.

3. Project Methodology

Even though I'm working alone on this project, I'll use the Scrum framework and Agile technique. I can construct the system using agile in reasonable chunks, test them repeatedly, and adjust as I go. I can concentrate on finishing one feature at a time, evaluate its functioning, and modify the following stages in response to results and input from supervisors or potential users when I work in short sprints.

Despite working alone, I will stick to Scrum's concepts by setting specific objectives for every sprint, keeping track of my to-do list, and giving features like patient registration, test scheduling, sample management, and report creation priority. This method guarantees organized development, lowers the possibility of mistakes, and enables me to create the system quickly while maintaining its adaptability to any modifications or improvements required for the lab management system.

3.1. Plan/Schedule

Being in proper format helps to complete the work in time. So I made following Gantt chart to help me complete the designed system in time.

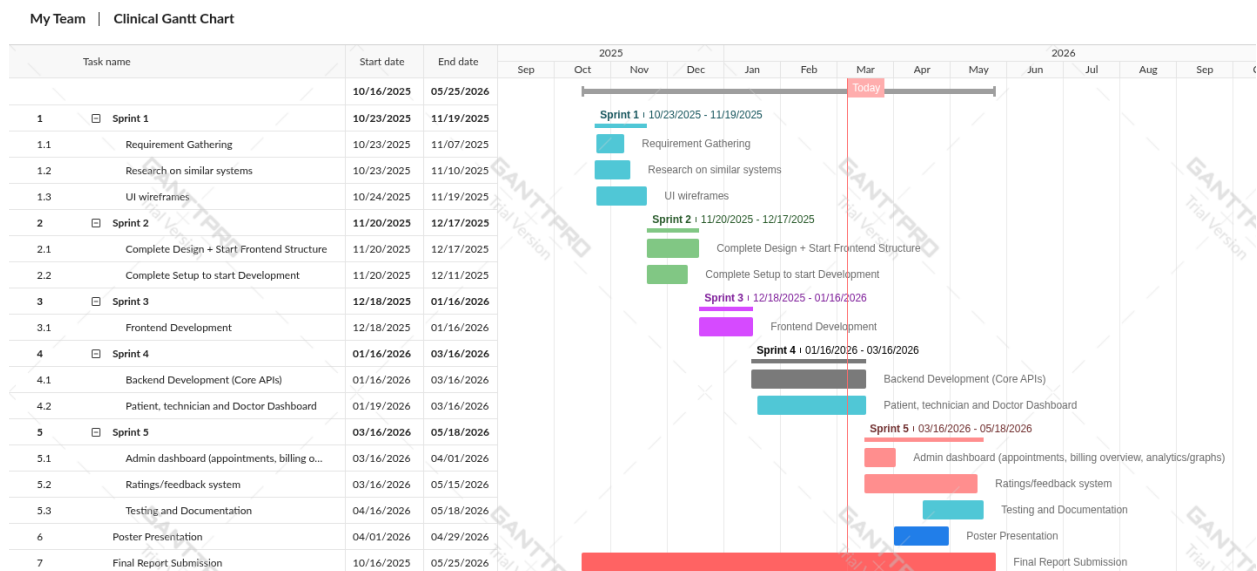


Figure 2 Gantt Chart

4. Different Technology and Tools used for the project

4.1. Tech Stack

For this project, I will use the **MERN stack** (MongoDB, Express.js, React.js, Node.js) along with supporting tools for UI design and version control. This stack allows full-stack JavaScript development, making it easier to maintain consistency and scalability.

- **Frontend: React.js** for a responsive, interactive, and user-friendly interface. Its component-based architecture simplifies managing complex pages like patient dashboards and report viewing.
- **Backend: Node.js** with **Express.js** to handle server-side logic and APIs. This combination ensures fast performance and smooth handling of multiple requests.
- **Database: MongoDB** for flexible, scalable storage of patient records, test details, and reports.
- **UI/UX Design: Figma** to create wireframes and prototypes, ensuring a clean and user-friendly interface.
- **Version Control: GitHub** to track changes, maintain code history, and organize development.
- **Hosting:** Platforms like **Vercel** (frontend) and **Render/Heroku** (backend) for easy deployment and online access.

Justification: Each tool in this stack is chosen because it provides better performance, flexibility, or ease of use compared to alternatives, ensuring a smooth, scalable, and user-friendly medical lab system.

4.2. Font style and icons

The user interface of the Clinical Medical Lab Management System uses modern web fonts to maintain readability and a clean visual design. The primary font used on the website is **Inter**, a widely used sans-serif font that provides good readability across different screen sizes and devices. It ensures that dashboard data, patient details, and reports are displayed clearly and consistently.

For visual elements and interface indicators, the system uses **Lucid React icons**, a lightweight and modern icon library designed for React applications. These icons are used to represent common actions such as user management, bookings, reports, notifications, and dashboard navigation. The use of icons improves the overall usability of the system by making features easier to identify and interact with.

The combination of a clean font style and intuitive icons help create a user-friendly interface that enhances the overall experience for patients, lab technicians, doctors, and administrators.

5. Artefact Design

5.1. Artefact Introduction

This artefact presents the complete design and development documentation for **Clinical**, a web-based Medical Lab Management System. The purpose of this artefact is to clearly explain how the system was planned, structured, and implemented, using appropriate software engineering models and diagrams. It includes key artefacts such as functional decomposition diagrams, use case diagrams, activity diagrams, and system descriptions that collectively illustrate the behavior of the system and interactions between different user roles.

The artefact acts as a central reference for understanding both the technical and functional aspects of the project. It demonstrates how user requirements were translated into system features and how design decisions were made to ensure usability, security, and scalability. Additionally, this documentation supports future maintenance and enhancement of the system by providing a clear overview of workflows, roles, and responsibilities. Overall, the artefact ensures transparency in the development process and reflects the application of standard software engineering principles throughout the project lifecycle.

5.2. Software Requirements Specification (SRS)

5.2.1 Patient Management Module

Functional Requirements

ID	Requirement
F1	Patients can register using name, email, citizenship number, and password.
F2	System validates unique citizenship number.
F3	Patients can log in using email and password.
F4	Patients can view and edit profile details.
F5	Patients can register for lab tests.
F6	Patients can view active tests and their status.
F7	Patients can view completed test results.
F8	Patients can securely log out.

Non-Functional Requirements

ID	Requirement
NF1	Citizenship numbers must be stored securely.
NF2	Authentication must use JWT tokens.
NF3	Patient UI must be beginner friendly.

5.2.2. Doctor Management Module

Functional Requirements

ID	Requirement
F1	Doctors can log in.
F2	Doctors can search patients using citizenship number.
F3	Doctors can request lab tests.
F4	Doctors can view patient results.
F5	Doctors can add medical remarks.
F6	Doctors can log out.

Non-Functional Requirements

ID	Requirement
NF1	Doctor dashboards must support fast search.
NF2	System must restrict medical data visibility appropriately.

5.3.3 Lab Technician Management Module

Functional Requirements

ID	Requirement
F1	Technicians can log in.
F2	Technicians can view sample queue.
F3	Technicians verify patient identity via citizenship number.
F4	Technicians collect and label samples.
F5	Technicians enter test results.
F6	Technicians can report incidents.
F7	Technicians can log out.

Non-Functional Requirements

ID	Requirement
NF1	Sample queue must update quickly.
NF2	Technician UI must handle many records efficiently.
NF3	Data integrity must be preserved.

5.4. Wireframing and Designing

Wireframing and design play an important role in building a successful software system because they help turn ideas into clear, visual plans before development begins. By using wireframes, I was able to outline the structure of each screen, decide where key features should appear, and understand how users would navigate through the system. This helped me identify usability issues early, saving time and effort during development. Designing the interfaces in **Figma** allowed me to create clean, consistent, and user-friendly layouts that align with the overall theme of *Clinical*. It also made it easier to focus on the user experience, ensuring that patients, lab technicians, doctors, and admins can interact with the system comfortably. Overall, wireframing and design helped me plan better, reduce errors, and build a more intuitive and professional-looking application.



Figure 3Wireframe Login Page

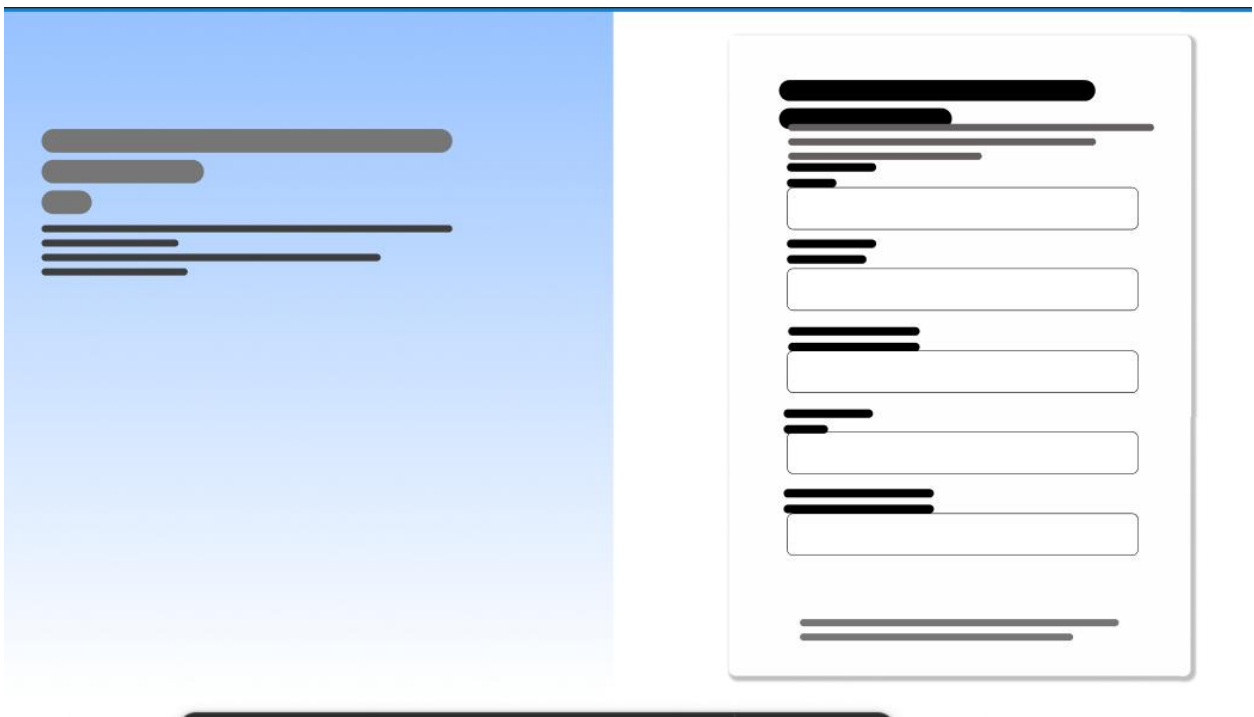


Figure 4Wireframe Registration Page

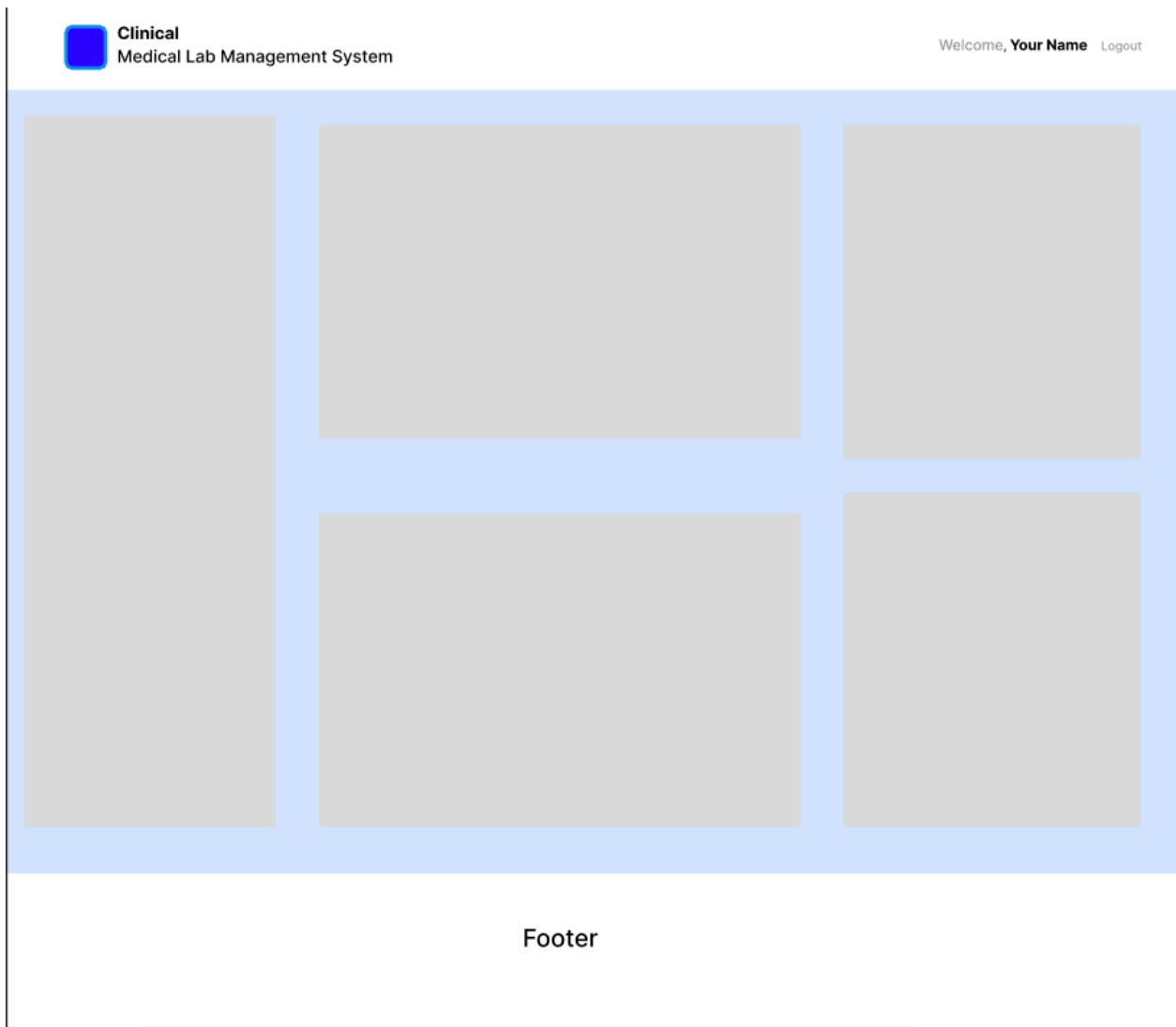


Figure 5 Wireframe Dashboard

Secure Lab Management

Welcome to Clinical

Manage patients, tests, and lab reports in one secure platform.
Fast, accurate, and designed for modern medical labs in Nepal.

Real-time tracking
Follow every sample from collection to report.

Secure records
Encrypted patient profiles with Citizenship ID.

Doctor access
Instant access to lab reports & notes.

Clinical – Medical Lab System

Sign in to Clinical

Access your lab dashboard, manage tests and medical reports.

Email *
you@example.com

Password *

☐ Keep me signed in [Forgot password?](#)

Sign in

Don't have an account? [Create an account](#)

Figure 6:Login Page

Patient registration

Create your Clinical account

Register once using your email and Citizenship ID to access lab tests, reports and consultations.

Clinical – Medical Lab System

Create your account

Fill the details below to register as a patient.

Full name *
Patient Name

Email *
you@example.com

Citizenship ID *
01-01-000000

Password *

Confirm password *

Create account

By creating an account you agree to Clinical's terms and privacy

Figure 7:User Registration Page

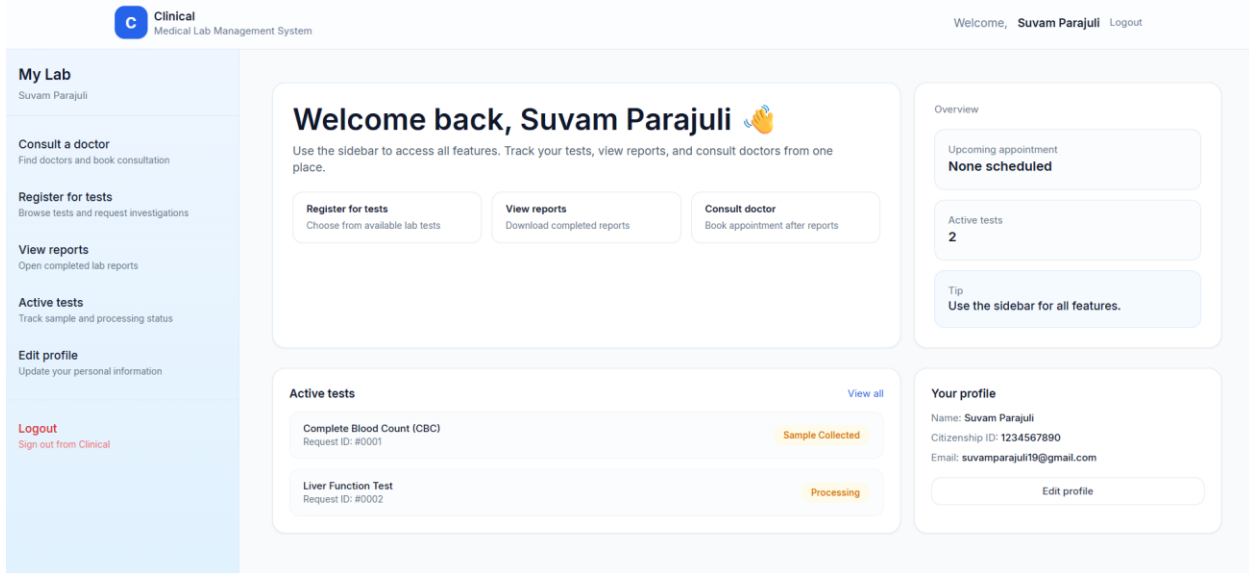


Figure 8: Patient Dashboard

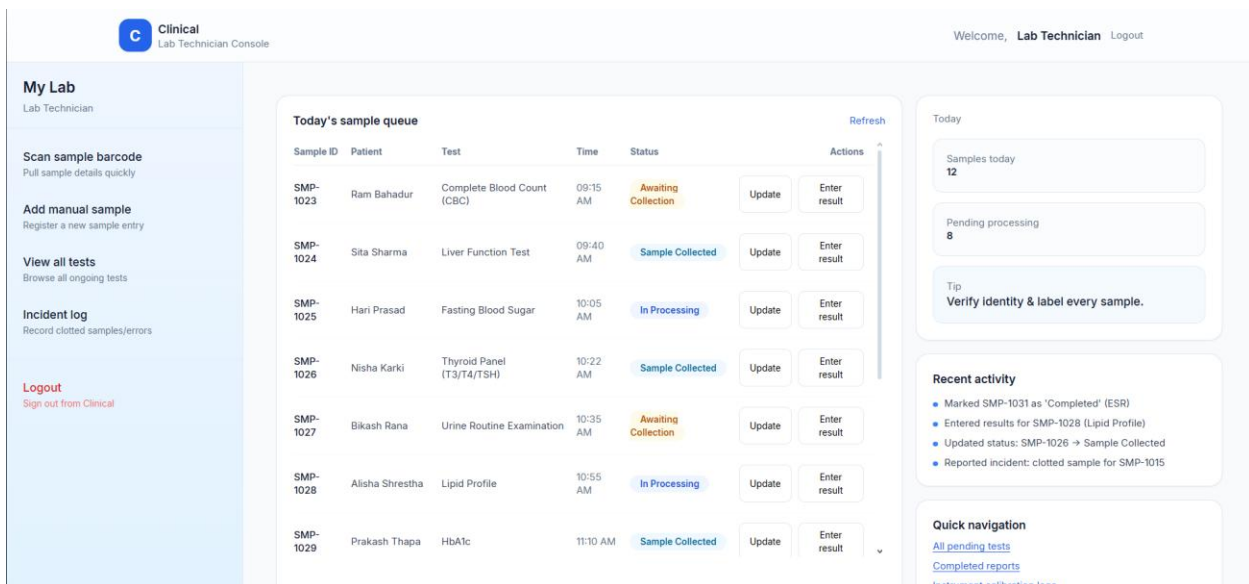


Figure 9: Technician Dashboard

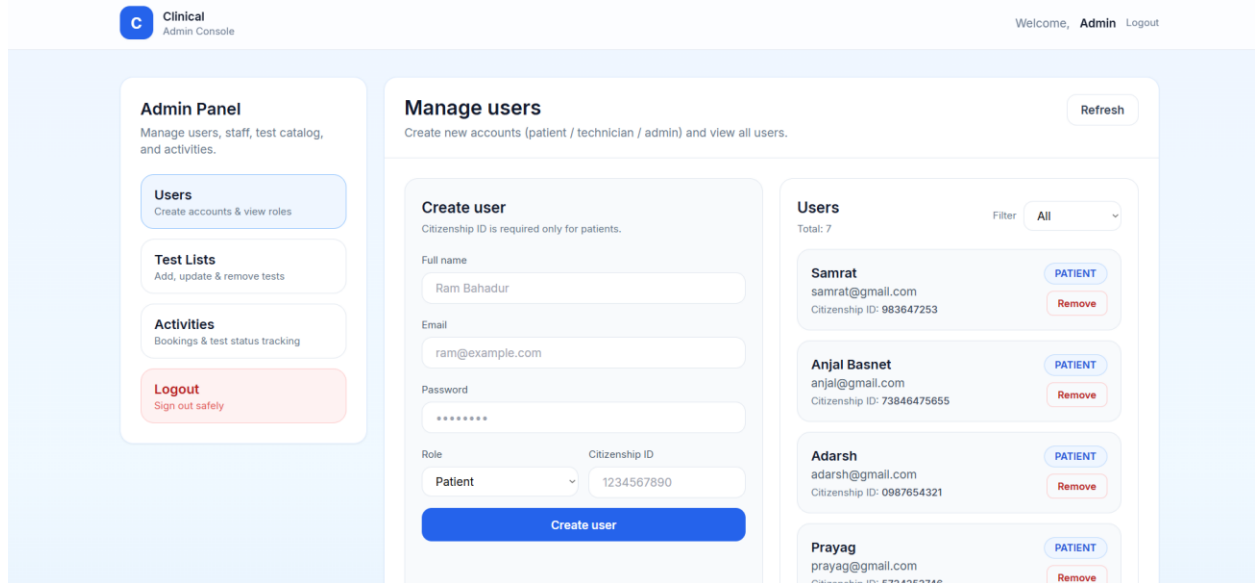


Figure 10:Admin Dashboard

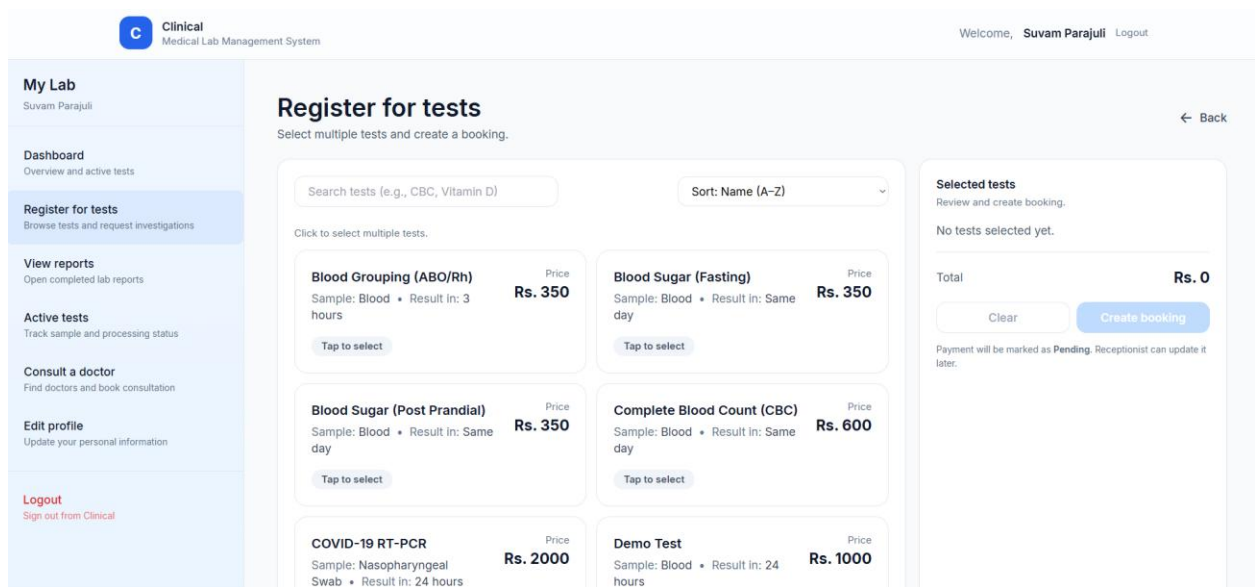


Figure 11:Register Test Page

5.5 Test Cases

5.5.1. User Module – Test Case

TC ID	Feature	Preconditions	Steps	Test Data	Expected Result
U-01	Register user	User not registered	Open Register → fill form → submit	name, email, citizenshipId, password	Account created, success message shown
U-02	Register with existing email	Email already exists in DB	Register using same email	existing email	Error: “Email is already registered”
U-03	Register password mismatch	None	Enter password ≠ confirm password → submit	pass1, pass2	Error: “Passwords do not match”
U-04	Login valid credentials	User exists	Open Login → enter email/password → submit	valid email/pass	Login success, token saved, redirect by role
U-05	Login invalid password	User exists	Enter correct email + wrong password → submit	wrong pass	Error: “Invalid email or password”
U-06	Logout	User logged in	Click Logout	—	Token/user removed, redirect to Login
U-07	Unauthorized dashboard access	Not logged in	Open /dashboard directly	—	Redirect to Login (RequireAuth works)

5.5.2. Patient Module – Test Case

TC ID	Feature	Preconditions	Steps	Test Data	Expected Result
P-01	View patient dashboard	Logged in as patient	Login → open /dashboard	patient user	Dashboard loads with patient name shown
P-02	Register for tests (basic)	Logged in as patient	Click Register for tests → select test → submit	testId(s)	Request created, confirmation shown
P-03	View active tests	Patients have active requests	Open Active tests section	—	List shows current tests with correct statuses
P-04	View reports available	Test completed + report exists	Open Reports → view/download	reportUrl	Report opens/downloads successfully
P-05	View reports not ready	Test not completed	Open Reports	—	Message: “Report not available yet”
P-06	Update profile (basic)	Logged in as patient	Click Edit profile → change info → save	name/email	Profile updated successfully (or success toast)
P-07	Patient cannot open technician page	Logged in as patient	Go to /technician	—	Redirect to /dashboard (role guard works)

5.5.3. Lab Technician Module – Test Case

TC ID	Feature	Preconditions	Steps	Test Data	Expected Result
T-01	Technician login	Technician user exists	Login with technician credentials	labtechnician@gmail.com / labtechnician	Redirect to /technician dashboard
T-02	View sample queue	Logged in as technician	Open Technician dashboard	—	Sample list/table loads correctly
T-03	Update sample status	Sample exists	Click Update → choose status	collected/processing/completed	Status updated and visible in queue
T-04	Enter result	Sample in processing	Click Enter result → submit values	resultSummary/values	Result saved, confirmation shown
T-05	Incident log entry	Sample exists	Open Incident log → submit issue	type + description	Incident saved and visible in activity/log
T-06	Technician cannot access admin page	Logged in technician	Go to /admin	—	Access denied/redirect (role guard works)
T-07	Technician logout	Logged in technician	Click Logout	—	Token/user removed, redirect to Login

5.6. UML Diagram

5.6.1. Activity Diagram

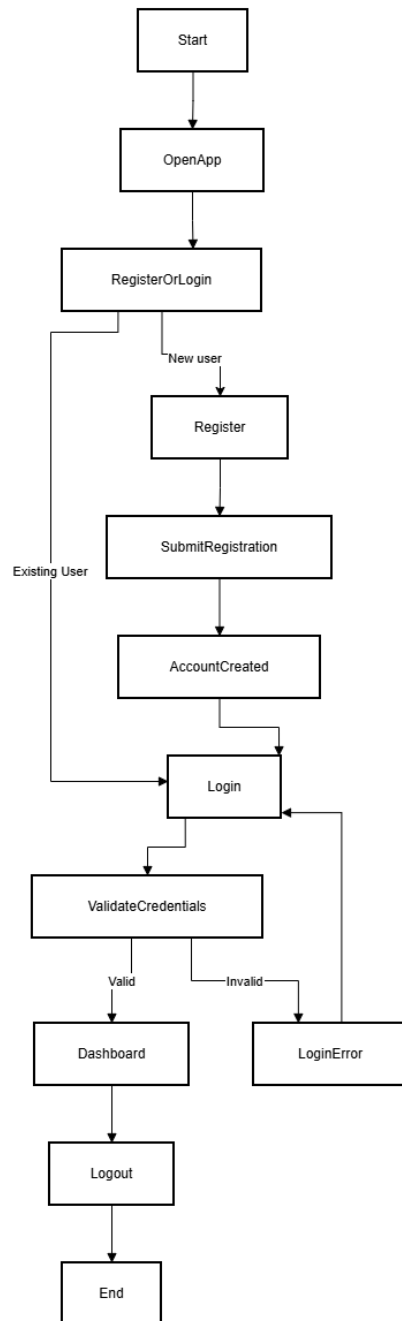


Figure 12: Activity Diagram of User

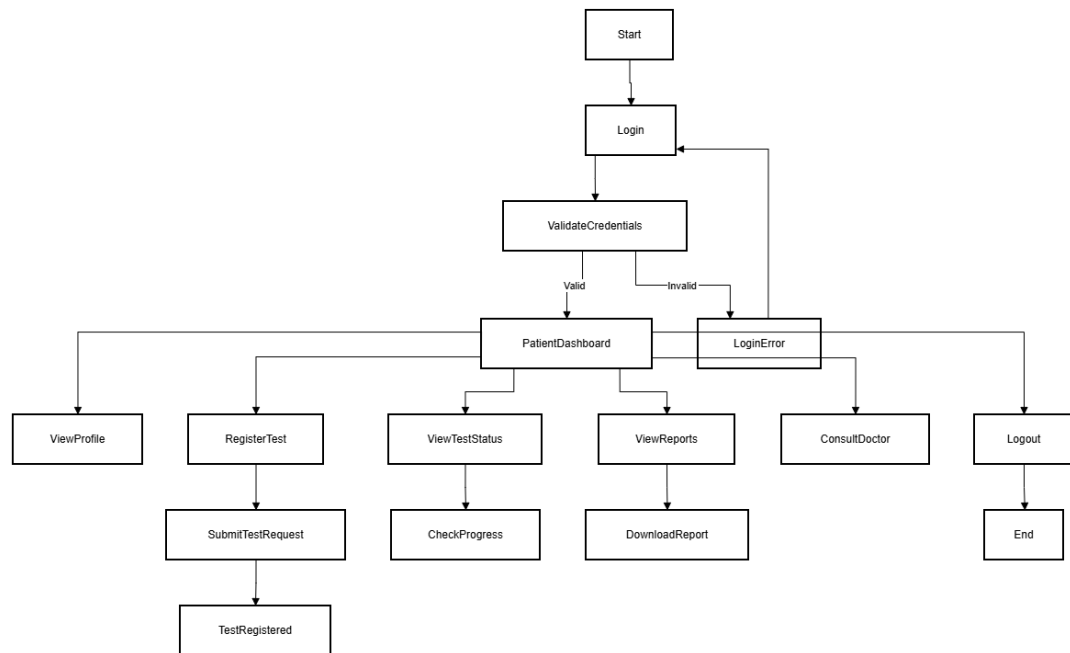


Figure 13:Activity Diagram of Patient

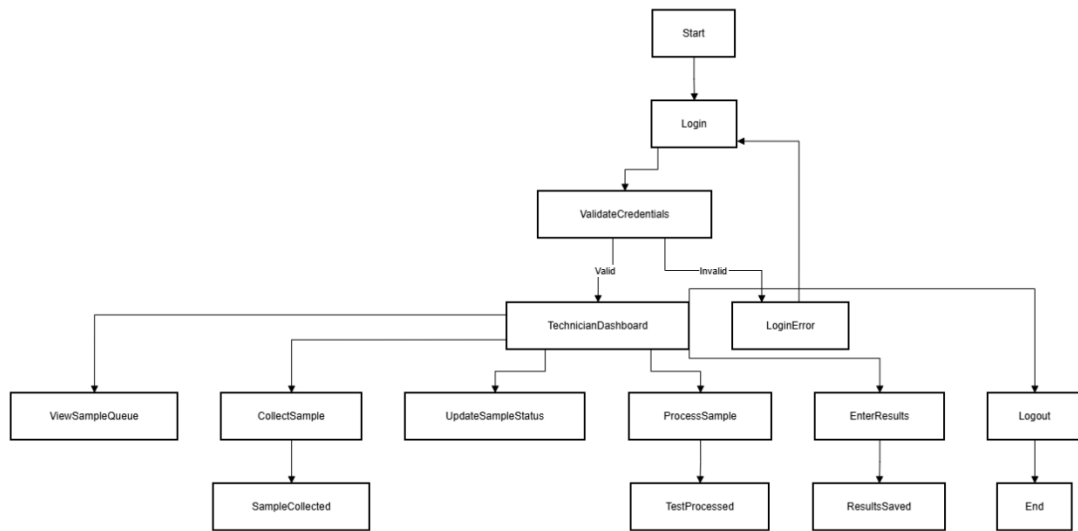


Figure 14:Activity Diagram of Technician

5.6.2. Use Case Diagram

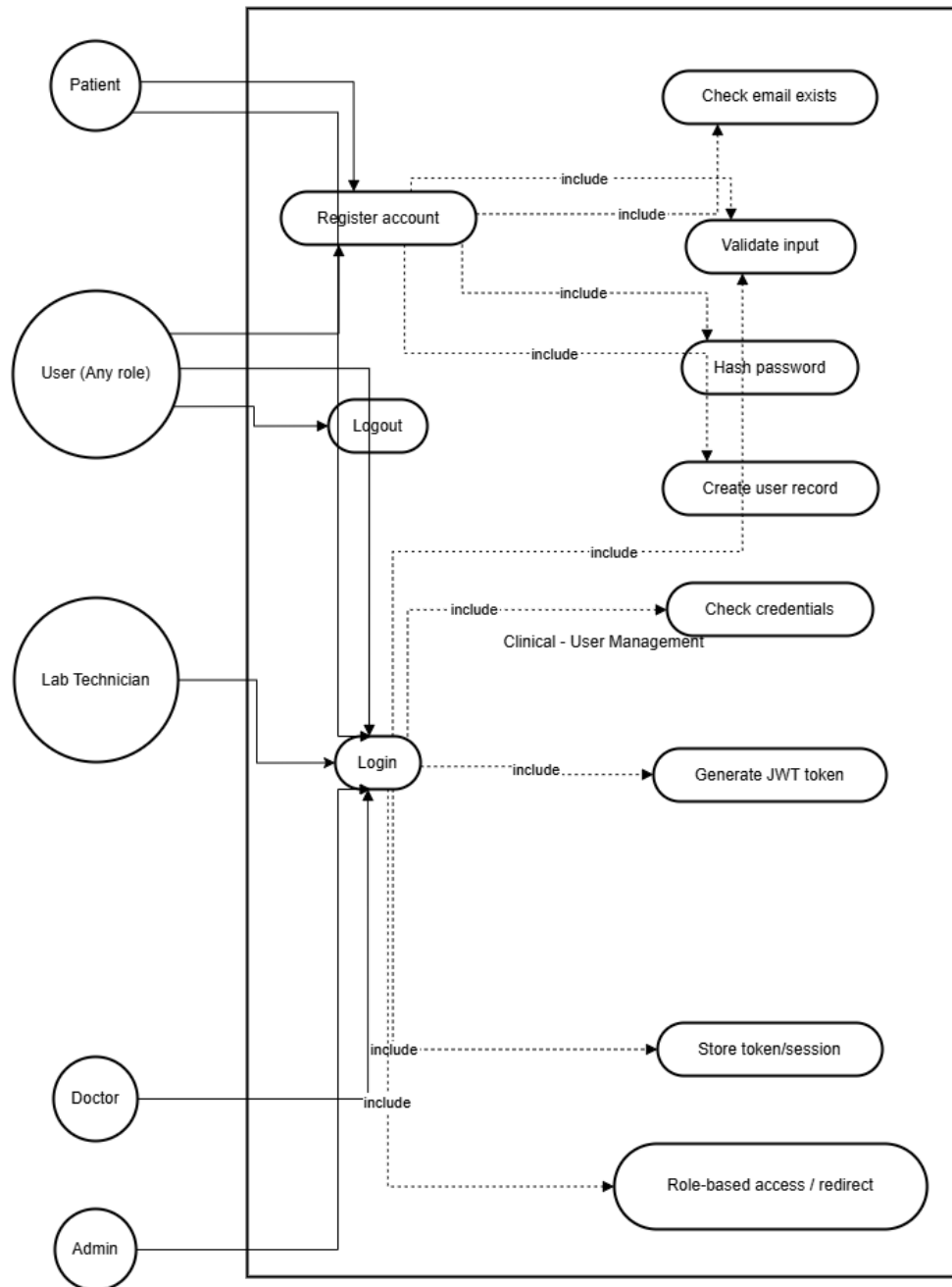


Figure 15: User's Use Case diagram

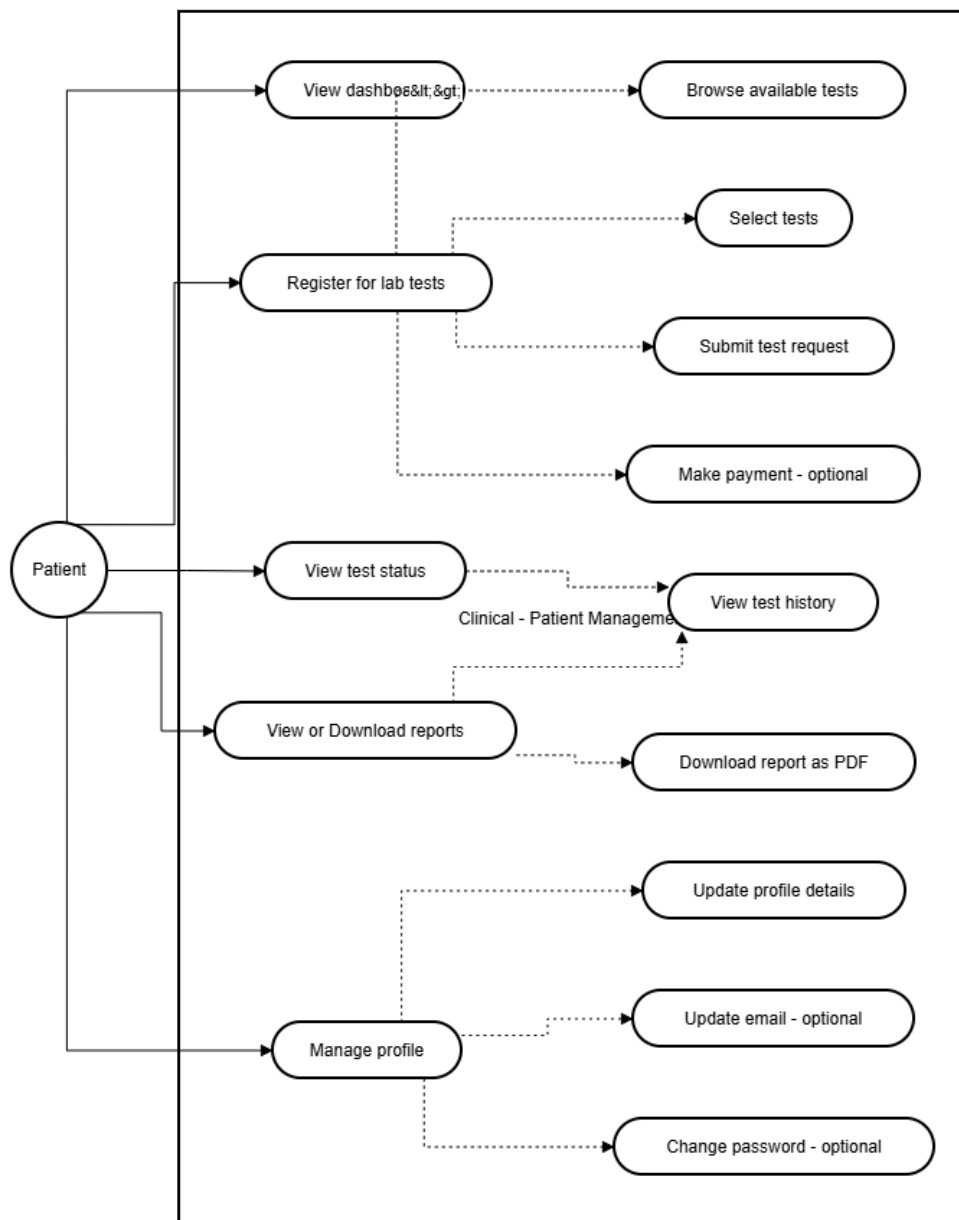


Figure 16: Patient's Use Case Diagram

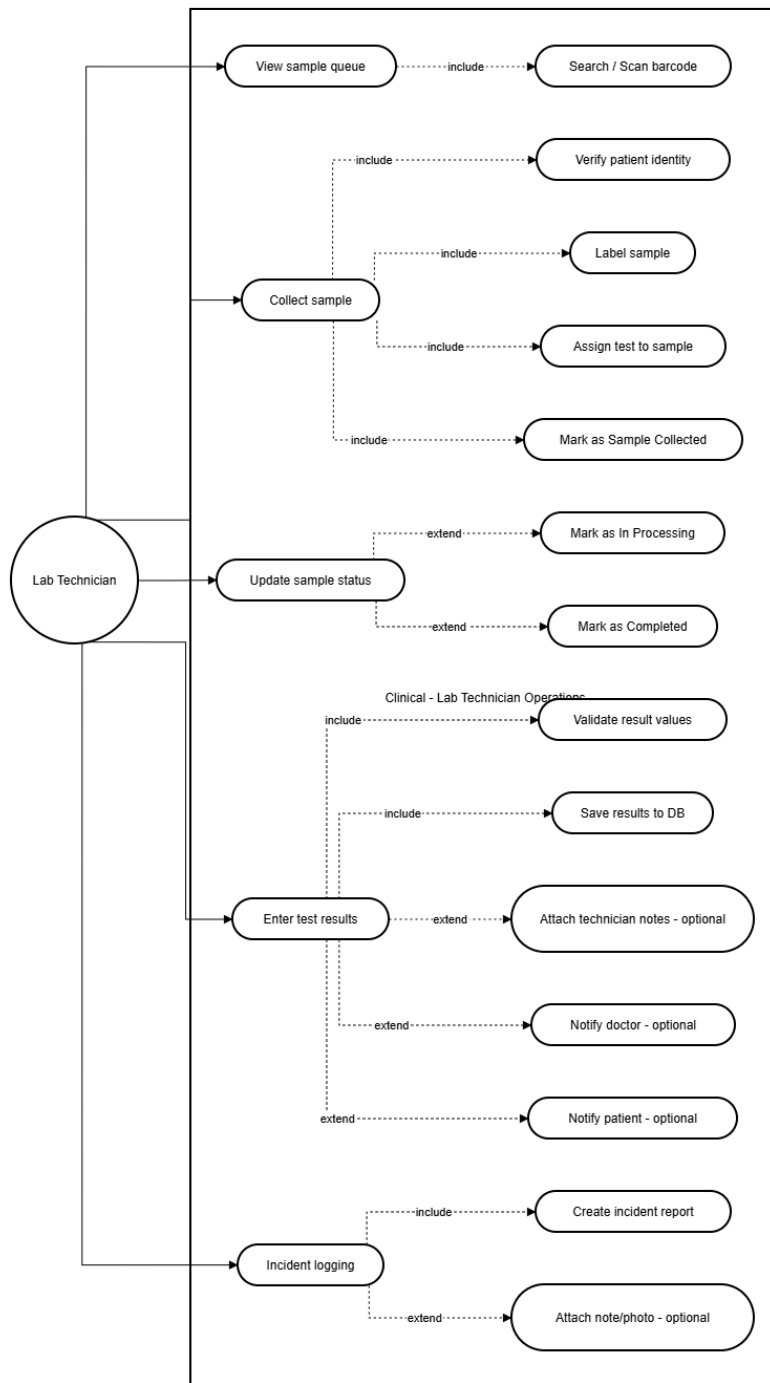


Figure 17: Technician's Use Case Diagram

5.6.3. Entity Relationship Diagram

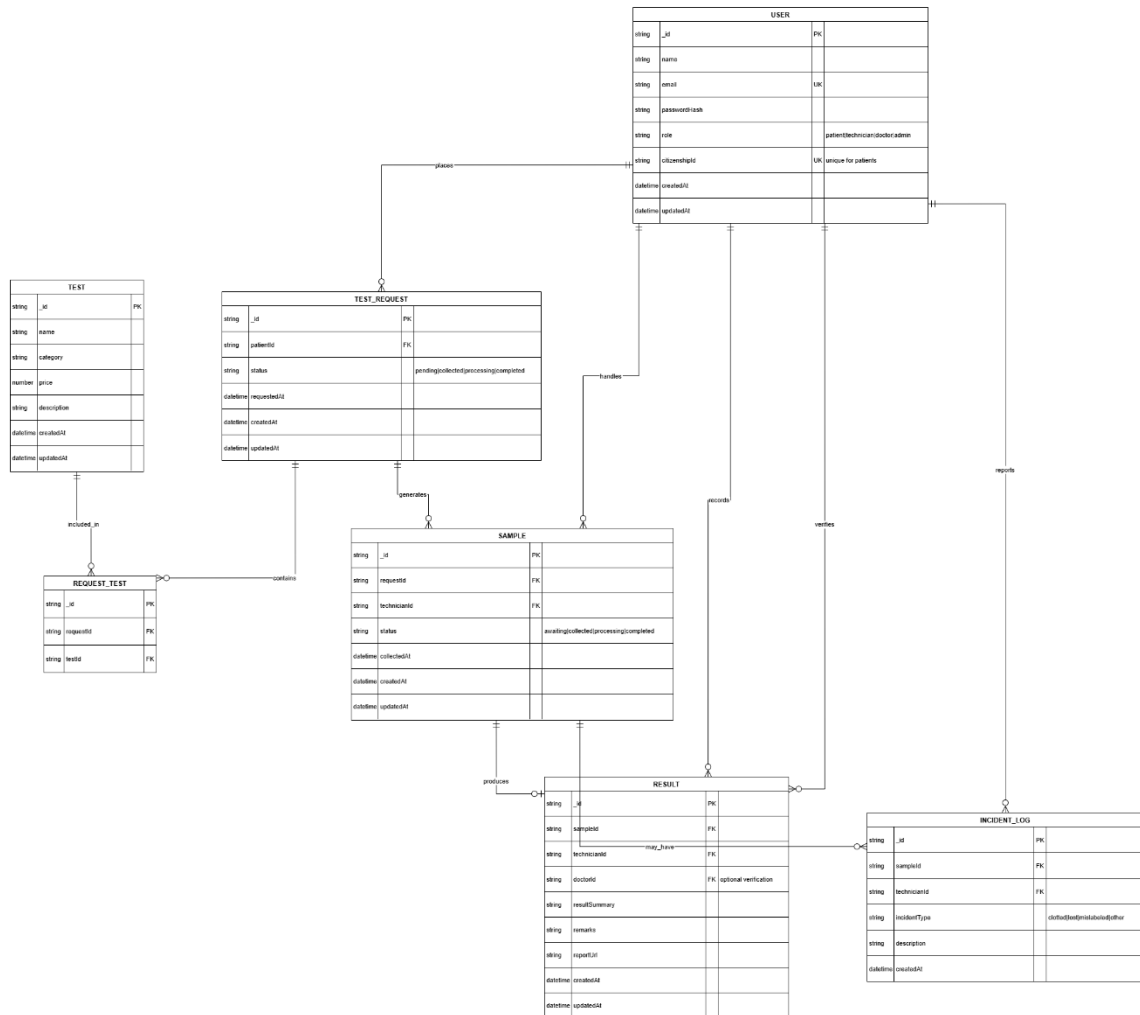


Figure 18: Entity Relationship Diagram

5.6.4. Class Diagram

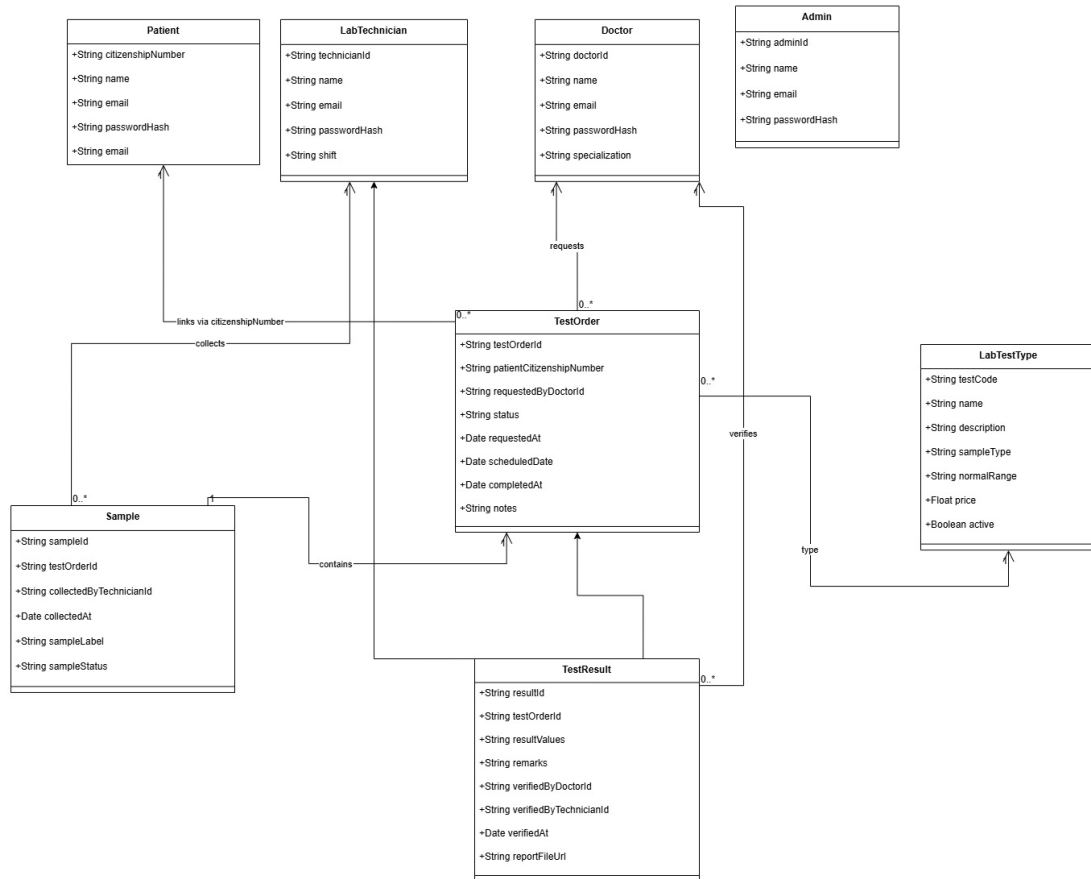


Figure 19: Class Diagram

5.6.5. Sequence Diagram

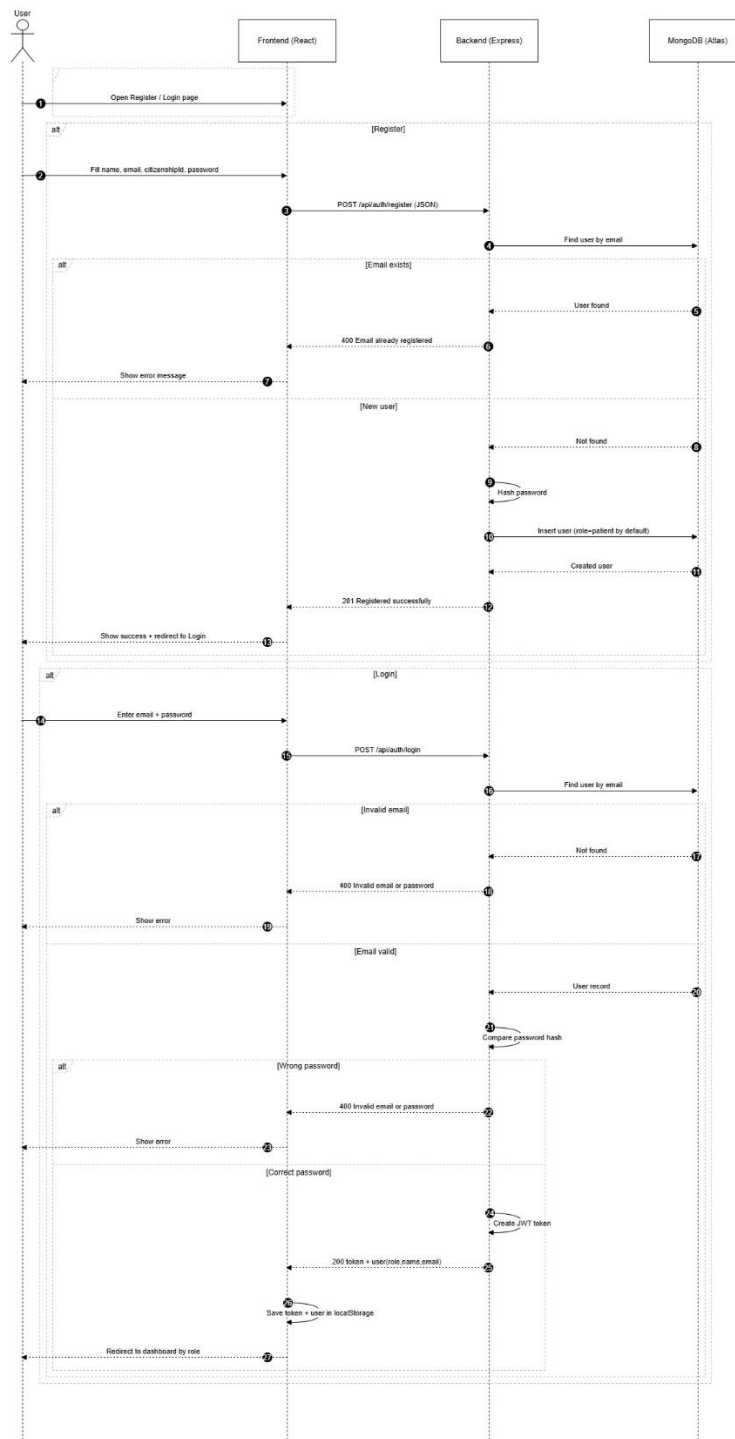


Figure 20: User Sequence Diagram

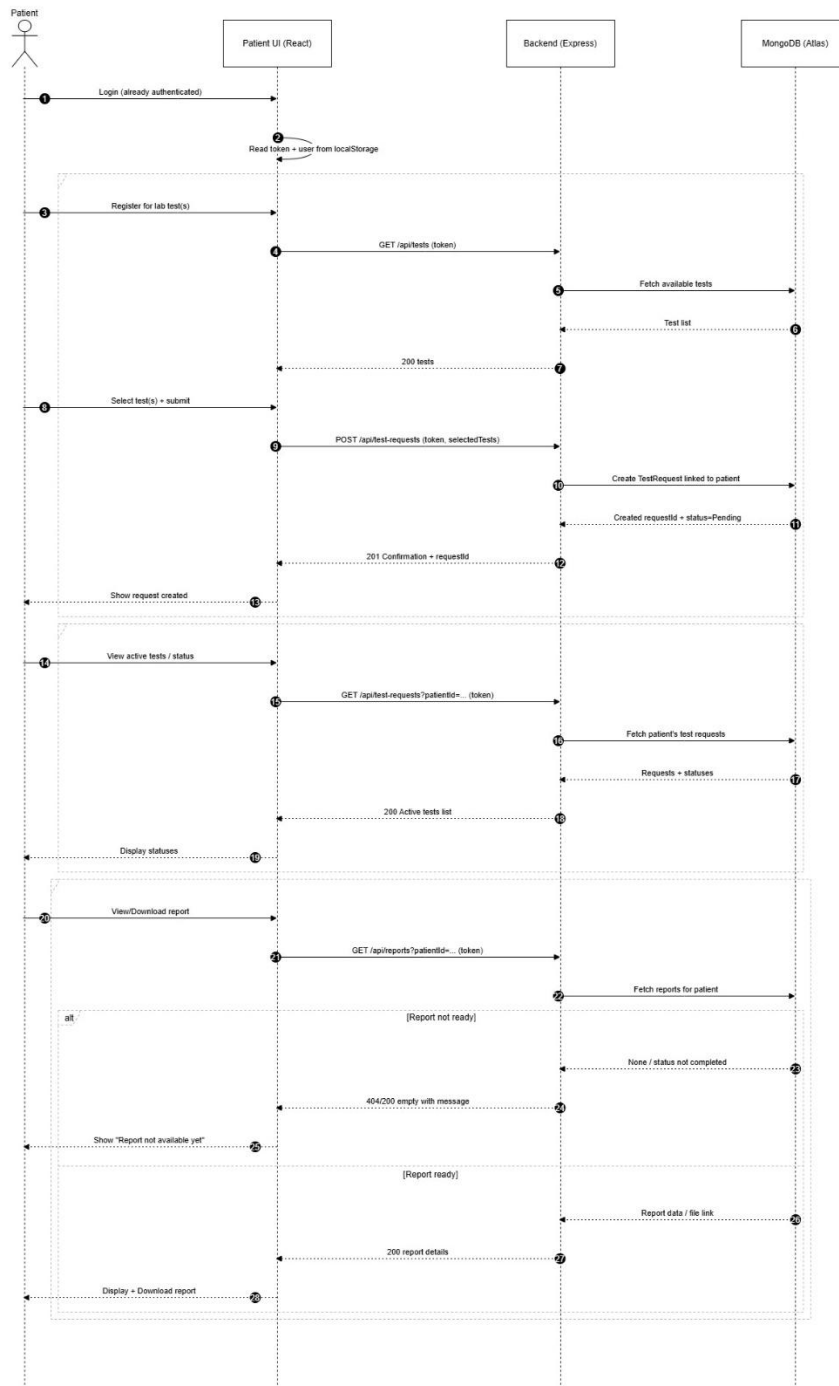


Figure 21: Patient Sequence Diagram

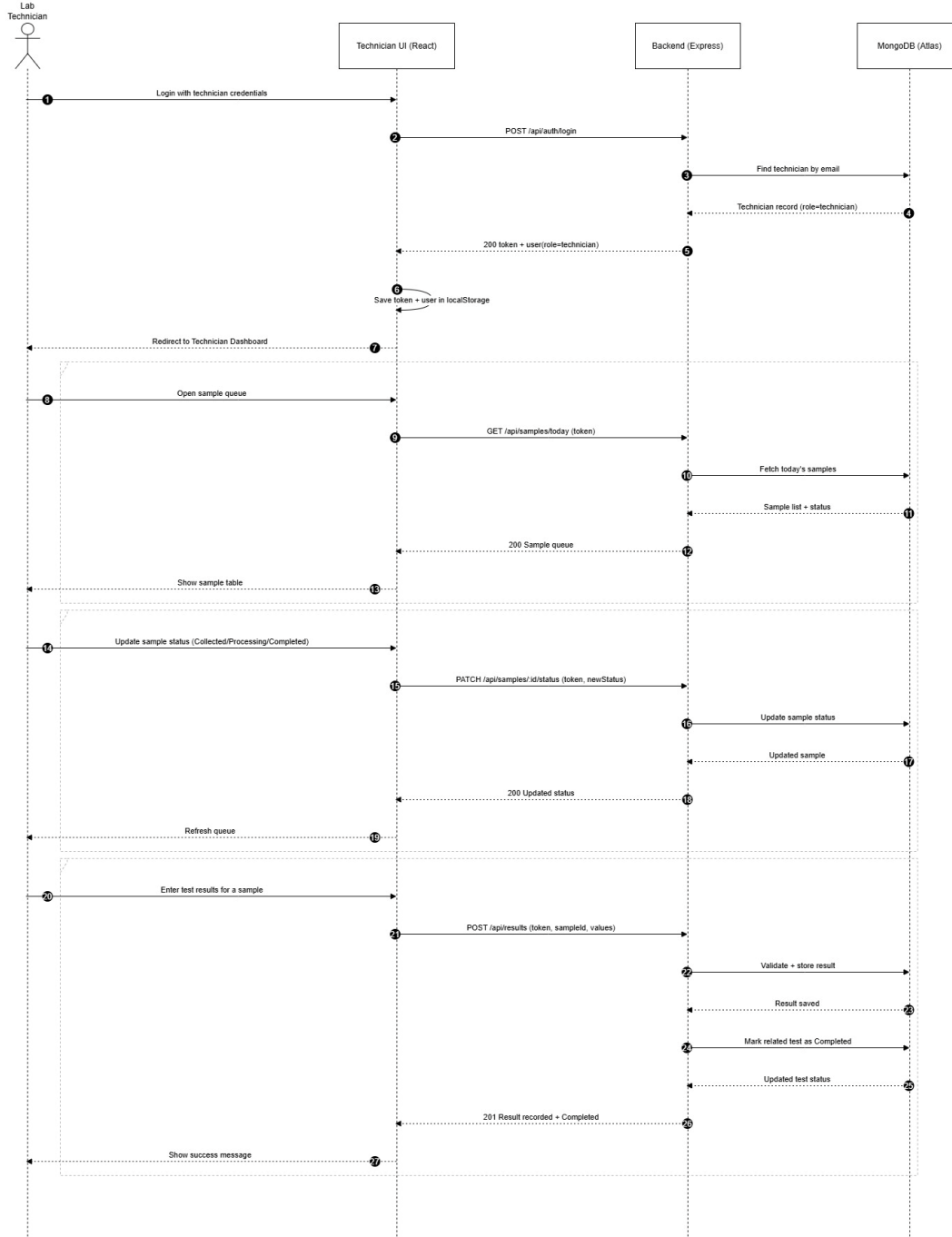


Figure 22: Technician Sequence Diagram

5.6.6. Conclusion

In conclusion, the artefacts designed for the **Clinical** system provide a clear and structured foundation for understanding and developing the application. Diagrams such as **use case diagrams, activity diagrams, ERD, FDD, and sequence diagrams** helped visualize system functionality, user interactions, and data flow across different modules. Wireframes and UI designs ensured a user-friendly and consistent interface for patients, lab technicians, and administrators. Together, these artefacts reduced ambiguity, improved communication among stakeholders, and guided accurate implementation. Overall, the artefact design phase played a vital role in ensuring that the system is well-planned, scalable, and aligned with project requirements.

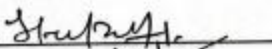
6. Log Sheets

Faculty of Science and Engineering
School of Mathematics and Computer Science

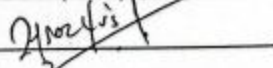
 UNIVERSITY OF
WOLVERHAMPTON

PROJECT MANAGEMENT LOG	
First Name: <u>Suvm</u>	Surname: <u>Parajuli</u>
Student Number: <u>2417725</u>	Supervisor: <u>Suraj Pandey</u>
Project Title: <u>Clinical</u>	Month: <u>November</u>
What have you done since the last meeting	
• Completed proposal. • Wireframes and UI part started.	
What do you aim to complete before the next meeting	
• Literature Review. • More UI designing.	
Supervisor comments	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

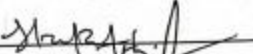
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Supervisor Signature: 

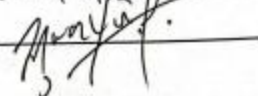
Date: 11/12/2025

PROJECT MANAGEMENT LOG	
First Name: <u>Suvam Parajuli</u>	Surname:
Student Number: <u>2417725</u>	Supervisor: <u>Suraj Pandey</u>
Project Title: <u>Clinical</u>	Month:
What have you done since the last meeting	
<u>I have completed literature review, some parts of UI design, wireframing.</u>	
What do you aim to complete before the next meeting	
<u>To complete the designing part by next meeting.</u>	
Supervisor comments	
<u>To start system development as well in parallel with the system design.</u>	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

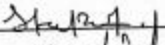
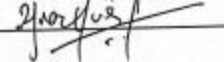
Date: 11/12/2025

Supervisor Signature: 

Date: 11/12/2025

PROJECT MANAGEMENT LOG	
First Name: <u>Suvam</u>	Surname: <u>Parajuli</u>
Student Number: <u>2417725</u>	Supervisor: <u>Suraj Pandey</u>
Project Title: <u>Clinical</u>	Month: <u>November</u>
What have you done since the last meeting	
• Did more UI part, started setup for development. • Literature review, proposal.	
What do you aim to complete before the next meeting	
Start developing and coding.	
Supervisor comments	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 
Supervisor Signature: 

Date: 11/16/2025
Date: 11/16/2025

PROJECT MANAGEMENT LOG	
First Name: <u>Siddhant</u>	Surname: <u>Parajuli</u>
Student Number: <u>2417225</u>	Supervisor: <u>Suraj Pandey</u>
Project Title: <u>Clinical</u>	Month: <u>November</u>
What have you done since the last meeting	
• Started frontend for login and registration page. • started frontend for dashboard.	
What do you aim to complete before the next meeting	
Complete the user dashboard, add more details and features.	
Supervisor comments	
Continue with the artefacts as well. Design/Develop the system (login, registration). Lab technician interface.	

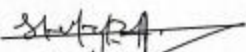
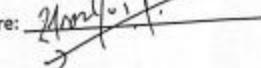
We confirm that the information given in this form is true, complete and accurate.

Student Signature: [Signature]
Supervisor Signature: [Signature]

Date: 11/23/2025
Date: 11/23/2025.

PROJECT MANAGEMENT LOG	
First Name: <u>Suvam</u>	Surname: <u>Parajuli</u>
Student Number: <u>2417725</u>	Supervisor: <u>Suraj Pandey</u>
Project Title: <u>clickd</u>	Month: <u>December</u>
What have you done since the last meeting	
• Entity Relation Diagram. • SRS • Lab technician dashboard.	
What do you aim to complete before the next meeting	
• More artifact design. • Further development of Doctor Dashboard.	
Supervisor comments	

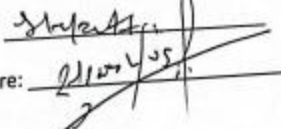
We confirm that the information given in this form is true, complete and accurate.

Student Signature: 
Supervisor Signature: 

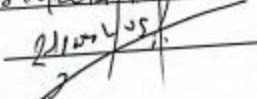
Date: 7/12/2025
Date: 7/12/2025

PROJECT MANAGEMENT LOG	
First Name: <u>Suvam</u>	Surname: <u>Parajuli</u>
Student Number: <u>2417725</u>	Supervisor: <u>Suraj Pandey</u>
Project Title: <u>Clinical</u>	Month: <u>December</u>
What have you done since the last meeting	
• I have added user authentication. • I have done more artefact designing. • I have made UI changes in both the patient and technician dashboard. • Created unique Id and password to access technician dashboard.	
What do you aim to complete before the next meeting	
• To complete more artefact designing. • To start the backend part in dashboard pages. • To complete the footer part of dashboards.	
Supervisor comments	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

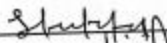
Date: 12/16/2025

Supervisor Signature: 

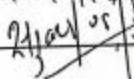
Date: 12/16/2025

PROJECT MANAGEMENT LOG	
First Name: <u>Suvam</u>	Surname: <u>Parajuli</u>
Student Number: <u>2417725</u>	Supervisor: <u>Suraj Pandey</u>
Project Title: <u>Clinical</u>	Month: <u>December</u>
What have you done since the last meeting	
• I have completed some works regarding the admin dashboard. • I changed the UI in multiple dashboard. • I have added footer in multiple pages.	
What do you aim to complete before the next meeting	
• Complete most of the part of artefact designing. • Upgrade the footer and admin dashboard. • Add test lists and prices.	
Supervisor comments	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

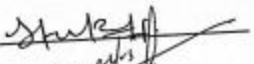
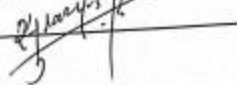
Date: 22/12/2025

Supervisor Signature: 

Date: 22/12/2025

PROJECT MANAGEMENT LOG	
First Name: Syvam	Surname: Parajuli
Student Number: 2417725	Supervisor: Suraj Pandey
Project Title: Clinical	Month: December
What have you done since the last meeting	
• Completed artefact designing. • Added test list with prices. • Added test booking feature for patient. • Updated Lab Technician and Admin dashboards. (They can view test list and test booked by patients).	
What do you aim to complete before the next meeting	
• Update test status from Lab Technician. • Admin can add/delete tests. • More necessary UI changes.	
Supervisor comments	

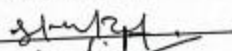
We confirm that the information given in this form is true, complete and accurate.

Student Signature: 
Supervisor Signature: 

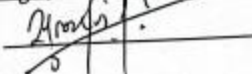
Date: 28/12/2025
Date: 28/12/2025

PROJECT MANAGEMENT LOG	
First Name: <u>Suvedn</u>	Surname: <u>Pardjuli</u>
Student Number: <u>2417725</u>	Supervisor: <u>Suraj Pandey</u>
Project Title: <u>Clinical</u>	Month: _____
What have you done since the last meeting	
• Made Update button from technician dashboard working were technicians can now change the process status. • Created different table where published test reports / tests are listed.	
What do you aim to complete before the next meeting	
• Make the enter result button working. • Results should be able to viewed by both user and technician. • Make the test result downloadable.	
Supervisor comments	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

Date: 1-4-2026

Supervisor Signature: 

Date: 1/4/2026

PROJECT MANAGEMENT LOG	
First Name: <u>Suvodh</u>	Surname: <u>Pardjuli</u>
Student Number: <u>2417725</u>	Supervisor: <u>Suraj Pandey</u>
Project Title: <u>Clinical</u>	Month: _____
What have you done since the last meeting	
• Lab technician and Patient can now view and download the published test result in pdf form. • Created downloadable report format.	
What do you aim to complete before the next meeting	
• Admin should also be able to look / track the active tests. • Make test result format more advanced.	
Supervisor comments	
* Work on test schedule or you can include it in register.	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: [Signature]

Date: 1-11-2026

Supervisor Signature: [Signature]

Date: 1-11-2026

First Name: <u>Suvam</u>	Surname: <u>Parajuli</u>
Student Number: <u>2417725</u>	Supervisor: <u>Kaushal Kishor Mishra</u>
Project Title: <u>Clinical</u>	Month: <u>February</u>

- Added search option in published tests table in technician dashboard.
- All the published test results can be viewed by technician.
- Added feature for admin from where CRUD operation of tests can be done.

- To add doctors and make dashboard for doctors.
- Add more features in doctor's dashboard.

We confirm that the information given in this form is true, complete and accurate.

Date: 21/22/2026

Date: 21/22/2026

7. References

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